

Examiner reports

Q1.

This question was a good source of marks for most students. In part (a), 95% of students correctly identified the outliers. Although some did not gain further marks, rather than offering an explanation as to the difference in Collins and Donovan's values compared to the others they just stated the reason they had picked them out as the outliers, effectively re-stating the co-ordinate values.

In part (b) students often lost the mark for not indicating the strength of the positive correlation.

Q3.

Most candidates scored at least 5 of the 7 marks available. In part (a), estimates were often sufficiently accurate in terms of magnitude, but a significant minority stated 'zero' or omitted the minus sign in part (b).

Q4.

Almost all candidates scored at least 2 marks on this question. Answers to part (a) were usually within the range necessary to score 2 marks; if not, they were within the wider range that scored 1 mark. Candidates were less confident in answering part (b), with a significant proportion of candidates quoting a value that indicated a strong correlation.

Q5.

Most, but not all, candidates scored the 2 marks available for each of parts (a) and (b).

In answering part (c), it was not sufficient to state as the reason that " $r > 0$ "; an indication that (for boys aged 11 to 18) weight was likely to increase with age was required.

Q6.

Statistics 1A

Fewer candidates than expected gained full marks in this question, usually due to a lack of precision in answers to part (a) and/or part (c).

In part (a), many candidates commented on a correlation between temperature and sales, although they were not always sufficiently specific to gain the mark. Adjectives such as 'weak positive' or descriptions suggesting 'a slight increase' were expected. Recognition of anomalies needed to be clear; mere statements of maximums, minimums or ranges of values, that could be obtained from the table, were not sufficient.

In part (b), most candidates correctly identified Monday 10. By far the most common error was stating Monday 4; candidates perhaps confused between Monday 4 and the 4°C from Monday 10?

In part (c), 'weather' was not sufficient since the question had suggested temperature as a variable affecting the sales. Many other alternatives were acceptable though 'rain' was by far the most common valid answer.

Fewer candidates than expected gained full marks in this question, usually due to a lack of precision in answers to part (a) and/or part (c). A number of centres had clearly not yet taught the product moment correlation coefficient.

Statistics 1B

In part (a), many candidates commented on a correlation between temperature and sales, although they were not always sufficiently specific to gain the mark. Adjectives such as 'weak positive' or descriptions suggesting 'a slight increase' were expected. Recognition of anomalies needed to be clear; mere statements of maximums, minimums or ranges of values, that could be obtained from the table, were not sufficient.

In part (b), most candidates correctly identified Monday 10. By far the most common error was stating Monday 4; candidates perhaps being confused between Monday 4 and the 4°C from Monday 10?

In part (c), 'weather' was not sufficient since the question had suggested temperature as a variable affecting the sales. Many other alternatives were acceptable though 'rain' was by far the most common valid answer.

Q7.

Most candidates obtained good marks on this question although a few gave answers such as 'negative correlation' rather than providing the estimates asked for.

In part (b) some ignored the fact that the correlation coefficient was clearly negative.

Very small answers to part (c) were not uncommon - these candidates failing to note that if the origin is moved to $(\bar{x}, \bar{y})$ then most points clearly lie in the 2nd or 4th quadrant. This indicates a substantial negative correlation coefficient.

Q8.

Many candidates considered only the size of the coefficient and ignored the fact that the coefficient should be negative. They therefore said statement 1 was plausible while correctly explaining that towns with a well qualified population were likely to have low unemployment. Many assumed that each of the classes A, B and C would fit exactly one statement. This was true for this question but may not always be the case. Those who made this assumption and who misclassified one of the statements were then compelled to wrongly classify another.

Q9.

This question, requiring an entirely descriptive answer, was the least well answered. Even so most candidates realised that the product moment correlation coefficient is not appropriate for non-linear relationships.

Q10.

Again, this question was well answered. Some candidates gave a numerically larger answer for part (b) than for part (a). Other candidates gave impossible answers but such answers were very much in the minority. The question asked for an estimate of the correlation coefficient so answers such as 'negative correlation' for part (a) were not

considered adequate.

Part (c) caused some problems, with '0' being a common answer. The easiest way to answer this type of question is to make a rough estimate of where the point $(\bar{x}, \bar{y})$ lies and then visualise (or sketch) axes with this point as origin. Candidates could then observe how the points are distributed between quadrants.

Q11.

In part (a)(i), some very strange incorrect versions of a scatter diagram were produced. In part (a)(ii), the language used to describe the association was not as rigorous as expected – use of positive/direct or negative/indirect/inverse with an indication of strong or weak is advised. Most candidates had a good attempt at part (b) but hypotheses were often incorrectly stated and SRCC tables were used in part (b).



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